

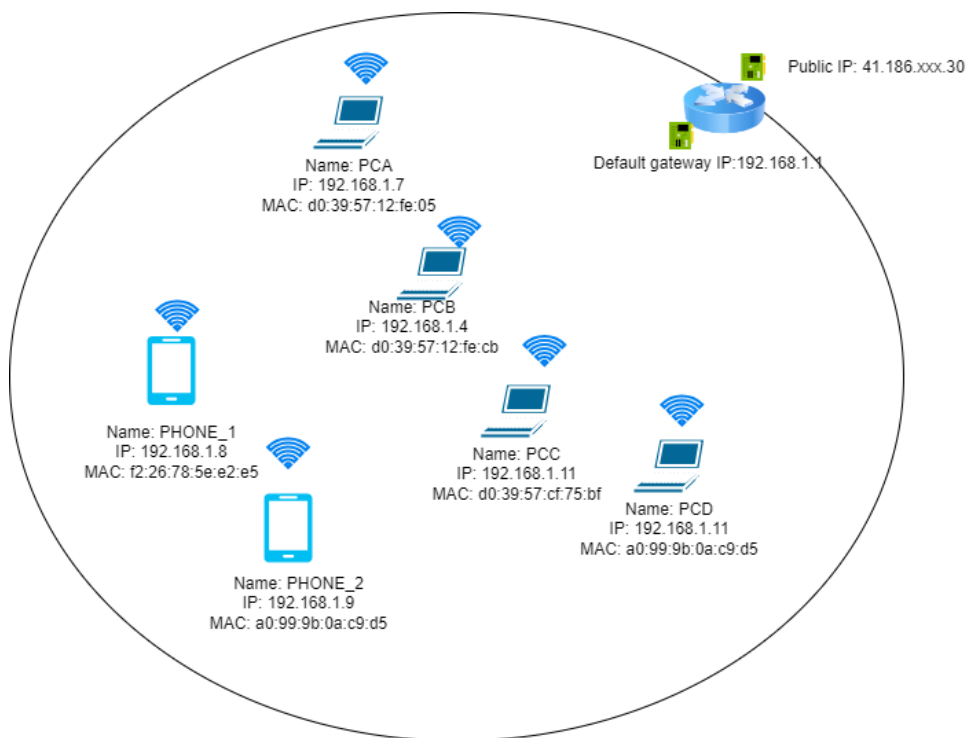
INTRO TO CYBER | PROJECT: NET CRAFTS

UNIT: RW-University-II

CODE: s10

1. Map the Network

1.1 up to 1.4 and 1.6



1.4. Display the DNS and DHCP IP Addresses in your Network.

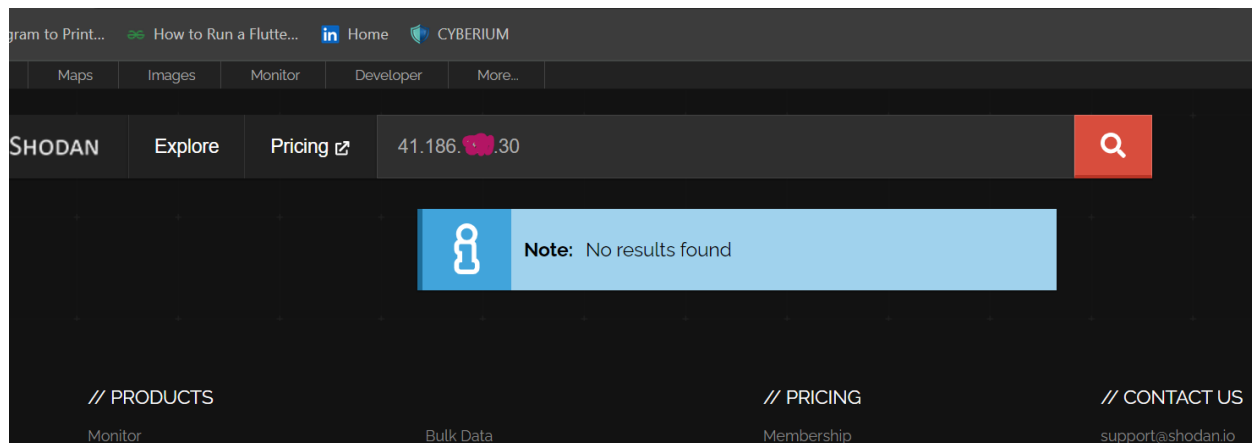
```
DHCP Server . . . . . : 192.168.1.1
DNS Servers . . . . . : 192.168.1.1
```

1.5. Display your Internet Service Provider (ISP)

inetnum: 41.186.0.0 - 41.186.255.255
netname: MTNRW-NET2
descr: MTN Rwandacell
country: RW
org: ORG-MTN1-AFRINIC

2. Collecting information

2.1. Use Shodan to Check Your Public IP Address



2.2. Use WHOIS to Check Who is Registered on Your Public IP Address

```
organisation: ORG-MTN1-AFRINIC
org-name: MTN Rwandacell
org-type: LIR
country: RW
```

2.3. Sniff Your Network and Identify Three (3) Used Protocols

2.3.1. For Each Protocol Explain Its Usage

2.3.2. For Each Protocol, Find the Used Port Number

Apply a display filter ... <Ctrl-/>						
No.	Time	Source	Destination	Protocol	Length	Info
182	13.280303	172.217.170.174	192.168.1.7	TCP	66	443 → 51556 [ACK] Seq=1 Ack=2 Win=261 Len=0 SLE=1
183	14.100172	170.114.52.30	192.168.1.7	TCP	54	443 → 51286 [ACK] Seq=1 Ack=1 Win=24 Len=0
184	14.100223	192.168.1.7	170.114.52.30	TCP	54	[TCP ACKed unseen segment] 51286 → 443 [ACK] Seq=
185	14.145915	192.168.1.7	172.217.170.163	TCP	55	51557 → 443 [ACK] Seq=1 Ack=1 Win=509 Len=1 [TCP
186	14.203212	172.217.170.163	192.168.1.7	TCP	66	443 → 51557 [ACK] Seq=1 Ack=2 Win=261 Len=0 SLE=1
187	14.470281	192.168.1.7	170.114.52.30	TCP	1474	51286 → 443 [ACK] Seq=1 Ack=2 Win=516 Len=1420 [T
188	14.470281	192.168.1.7	170.114.52.30	TLSv1.2	1056	Application Data
189	14.470475	192.168.1.7	170.114.52.30	TLSv1.2	93	Application Data
190	14.482680	192.168.1.7	216.58.223.78	UDP	71	59022 → 443 Len=29
191	14.601811	192.168.1.7	216.58.223.78	UDP	71	59022 → 443 Len=29
192	14.620945	216.58.223.78	192.168.1.7	UDP	68	443 → 59022 Len=26

UDP, TCP, TLSv1.2

1. UDP (User Datagram Protocol):

- **Usage:** UDP is a connectionless protocol that provides a simple and lightweight method for transmitting data between hosts on a network. It is often used for

applications where real-time communication and speed are crucial, such as VoIP (Voice over Internet Protocol), online gaming, DNS (Domain Name System) resolution, and streaming media.

- **Used Port Number:** Port 443 is typically associated with TCP, not UDP. However, UDP can also use port 443 for specific applications, such as UDP-based VPN protocols like DTLS (Datagram Transport Layer Security).

2. TCP (Transmission Control Protocol):

- **Usage:** TCP is a connection-oriented protocol that ensures reliable, ordered, and error-checked delivery of data packets between devices over a network. It is widely used for applications that require guaranteed delivery of data, such as web browsing, email transmission (SMTP, POP3, IMAP), file transfer (FTP), and remote access (SSH, Telnet).
 - **Used Port Number:** Port 443 is commonly used for secure HTTP (HTTPS) connections over TCP, which encrypts data using TLS (Transport Layer Security) or its predecessor, SSL (Secure Sockets Layer). TLSv1.2 is a specific version of the TLS protocol used for securing communications over TCP connections, including HTTPS.
3. In Wireshark, seeing "application data" for TLSv1.2 indicates that data is being transmitted securely over a TCP connection using the TLS v1.2 protocol. This could include encrypted web traffic, secure email transmission, or any other application that relies on TLS for secure communication.